

EEG Session Quality Control Report

EFRCourierOpenLoop — BIDS Format

Purpose: This notebook implements a standardized, session-level quality control (QC) framework for intracranial EEG (iEEG) studies that combine cognitive task performance with direct brain stimulation. The pipeline is designed to run after each testing session to enable rapid corrective action if any errors are detected.

Experiment: *EFRCourierOpenLoop* — a spatial memory task in which patients navigate a virtual town and deliver packages to different stores, while intracranial brain stimulation is delivered at 3 Hz or 8 Hz theta-burst frequencies during encoding and/or retrieval phases.

Pipeline structure:

Section	Module	What it checks
0	Data Loading	BIDS file paths, channel info, event tables
1	Protocol Validation	Event counts, stimulation delivery, trial phasing
2	Behavioral Summary	Navigation time, spatial trajectory, recall performance
3	Stimulation EEG	Evoked responses at stimulated vs. non-stimulated contacts
4	Channel QC	Variance-based bad-channel detection, line-noise screening

How to use: Set `subject`, `session`, and `bids_root` in Section 0, then run all cells top to bottom. Figures are saved as PDFs to `write_fig_path` (default: current directory).

Section 0 — Data Loading

This section establishes the BIDS directory structure and loads three data sources that are used throughout the notebook:

1. **Raw iEEG signal** (`raw`) — loaded via `mne-bids` from the bipolar-referenced EDF file. This provides the continuous voltage time series for all channels.
2. **Channel metadata** (`channel_info`) — a BIDS TSV file containing bipolar pair names (`lead1`, `lead2`), electrode group (shank identity), and anatomical labels for every recorded channel.

3. Event tables — merged from two sources:

- `ieeg/..._events.tsv` — events synchronized to the EEG recording (onset in seconds relative to recording start).
- `beh/..._events.tsv` — behavioral events logged by the task computer, which may contain richer trial metadata (store names, item identities, stim conditions).

The two tables are concatenated and sorted by `onset` to produce `df_combined_sorted`, a unified chronological event log used by all downstream sections.

Key BIDS path parameters:

- `subject` — patient identifier (e.g., `'M1017'`)
- `session` — session number as a zero-padded string (e.g., `'0'`)
- `acquisition` — referencing scheme: `'bipolar'` (default) or `'monopolar'`

```
Checking BIDS root: /oceanus/collab//herz-lab/processed_data/bids/EfrCourierOpenLoop
✓ BIDS root found!
✓ Full path: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop
```

```
Loading file: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop/sub-M10
17/ses-0/ieeg/sub-M1017_ses-0_task-EfrCourierOpenLoop_acq-bipolar_ieeg.edf
Checking BIDS root: /oceanus/collab//herz-lab/processed_data/bids/EfrCourierOpenLoop
✓ BIDS root found!
✓ Full path: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop
```

```
Checking BIDS root: /oceanus/collab//herz-lab/processed_data/bids/EfrCourierOpenLoop
✓ BIDS root found!
✓ Full path: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop
```

```
Checking BIDS root: /oceanus/collab//herz-lab/processed_data/bids/EfrCourierOpenLoop
✓ BIDS root found!
✓ Full path: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop
```

```
Loading file: /oceanus/collab/herz-lab/processed_data/bids/EfrCourierOpenLoop/sub-M10
17/ses-0/ieeg/sub-M1017_ses-0_task-EfrCourierOpenLoop_acq-bipolar_ieeg.edf
```

Section 1 — Protocol Validation

The goal of this section is to verify that the session was executed as intended before data is analyzed. Protocol deviations such as truncated lists, missed stimulation pulses, or incomplete recall periods are identified.

1.1 Event Count Verification

The `EXPECTED_COUNTS` dictionary encodes the *a priori* expected number of each event type for a complete session. The check iterates over all event types and compares observed counts against

expected values, printing a color-coded PASS/FAIL table. Expected counts may vary across sessions; experimenters should adjust the expected count accordingly.

Event types and their meaning:

Event type	Meaning
session_start	Start of the entire session
trial_start	Start of a delivery day (encoding list) — 5 trials expected
encoding_word_start	A store visit where an item was encoded — 56 total
stimming	A delivered stimulation burst — 84 expected
pointing_start / pointing_stop	Spatial pointing test events — 69 each
fr_start / fr_stop	Free recall period boundaries — 5 each

=== M1017_0 ===

Event Type	Actual	Expected	Status
session_start	1	1	PASS ✓
trial_start	5	5	PASS ✓
encoding_word_start	56	56	PASS ✓
stimming	84	84	PASS ✓
pointing_start	69	69	PASS ✓
pointing_stop	69	69	PASS ✓
fr_start	5	5	PASS ✓
fr_stop	5	5	PASS ✓

8/8 checks passed

1.2 Stimulation Delivery Validation & Trial Phasing

This cell does three things:

Step 1 — Propagate burst frequency to adjacent events. When a `stimming` event is followed within 3 seconds by another event (e.g., `encoding_word_start`), the `burst_slow_frequency` from the stimulation event is copied forward. This ensures encoding-word events inherit the correct frequency label for downstream condition-tagging.

Step 2 — Assign trial numbers. The raw events table uses `trial_num` inconsistently (NaN outside of trial blocks). This step identifies `trial_start` boundaries, then fills `trial_num` sequentially so that every event row belongs to a uniquely identified delivery day.

Step 3 — Label task phases. The `add_phase()` function assigns each event a `phase` label (`'encoding'` or `'retrieval'`) based on its position relative to the `trial_start` →

`deliveries_stop` and `fr_start` → `fr_stop` event pairs within each trial. This phase label is essential for distinguishing stimulation effects during memory formation vs. memory retrieval.

Stimulation summary table (`parameters_df2`): For each trial, the table shows:

- Burst frequency (3 Hz or 8 Hz)
- Number of stimulation pulses during encoding and retrieval
- Number of items presented and correctly recalled
- Stimulation channel identity (derived by cross-referencing electrode contacts in `channel_info`)

1.2 STIMULATION DELIVERY VALIDATION

Stimulation distribution by phase :
Encoding – 3 Hz: 12, 8 Hz: 12, None: 29
Retrieval – 3 Hz: 30, 8 Hz: 30

	Burst Frequency (Hz)	Delivery Day	Number of Encoding Stim	Number of Retrieval Stim (EFR)	Number of Items Sent	Recalled Items	Stim Channel
0	[]	-999.0	0	0	0	[]	[]
1	[]	-1.0	0	0	8	[]	[]
2	[8.0]	0.0	6	15	12	3	RB1-RB2
3	[3.0]	1.0	6	15	12	3	RB1-RB2
4	[8.0]	2.0	6	15	12	2	RB1-RB2
5	[3.0]	3.0	6	15	12	2	RB1-RB2

Section 2 — Behavioral Summary

This section quantifies cognitive task performance through three complementary analyses: navigation efficiency, spatial trajectory, and memory recall accuracy. These behavioral metrics are meant to characterize the patient's engagement with the task and flag potential issues.

2.1 Navigation Time Analysis

Rationale: In the EFRCourierOpenLoop task, participants navigate a virtual town. The time elapsed between the patient's last pointing response at a store (`pointing_stop`) and the moment they reach that store to deliver an item (`encoding_word_start`) reflects how efficiently they are navigating the virtual environment.

`load_and_process_bids_events()` :

1. Extracts all `encoding_word_start` events and assigns each a stimulation condition (3 Hz, 8 Hz, or -999 = no stim) by finding the nearest `stimming` event.
2. For each encoded word, finds the most recent `pointing_stop` at the same store within the same trial — this is the moment the participant finished pointing to that location.
3. Computes `time = word_onset - pointing_stop_onset` (in seconds) as the navigation duration.

`plot_navigation_time()` : Produces a 2-panel figure:

- **Left panel:** Scatter plot of navigation time vs. trial number, color-coded by stimulation condition, with same-store trajectories connected by gray lines to reveal store-specific learning effects.
- **Right panel:** Histogram with median and mean values overlaid, annotated with skewness and kurtosis statistics.

Prolonged navigation times may indicate fatigue, inability to remember the town layout, or task interruptions, and should be flagged for review.

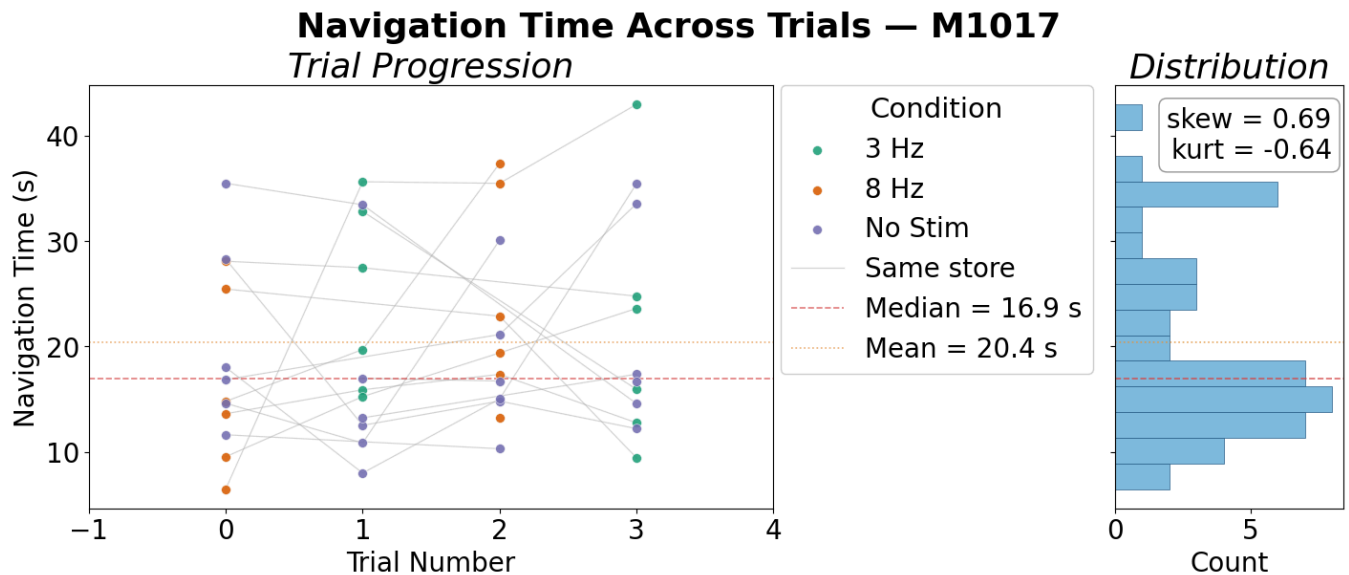
store	time	condition	visit_time	trial	session	subject
pharmacy	25.450455	8.0	0	0.0	0	M1017
pharmacy	22.876308	8.0	1	2.0	0	M1017
pharmacy	9.450541	3.0	2	3.0	0	M1017
bike shop	13.626779	8.0	0	0.0	0	M1017
bike shop	15.900909	3.0	1	1.0	0	M1017
bike shop	17.359993	8.0	2	2.0	0	M1017
bike shop	12.768730	3.0	3	3.0	0	M1017
grocery store	11.631665	-999.0	0	0.0	0	M1017
grocery store	10.309590	-999.0	1	2.0	0	M1017
bakery	28.084606	8.0	0	0.0	0	M1017
bakery	27.467571	3.0	1	1.0	0	M1017
bakery	24.773417	3.0	2	3.0	0	M1017
dentist	9.550546	8.0	0	0.0	0	M1017
dentist	15.251872	3.0	1	1.0	0	M1017
dentist	19.394109	8.0	2	2.0	0	M1017
dentist	23.607350	3.0	3	3.0	0	M1017
barber shop	14.617836	-999.0	0	0.0	0	M1017
barber shop	10.850620	-999.0	1	1.0	0	M1017
barber shop	30.075720	-999.0	2	2.0	0	M1017
pizzeria	14.792846	8.0	0	0.0	0	M1017
pizzeria	19.714128	3.0	1	1.0	0	M1017
pizzeria	37.360137	8.0	2	2.0	0	M1017
music store	35.476029	-999.0	0	0.0	0	M1017
music store	33.435912	-999.0	1	1.0	0	M1017
music store	14.589834	-999.0	2	3.0	0	M1017
hardware store	28.276618	-999.0	0	0.0	0	M1017
hardware store	12.526716	-999.0	1	1.0	0	M1017
hardware store	14.824848	-999.0	2	2.0	0	M1017
hardware store	12.211699	-999.0	3	3.0	0	M1017
florist	16.859964	-999.0	0	0.0	0	M1017
florist	21.136209	-999.0	1	2.0	0	M1017
florist	33.533918	-999.0	2	3.0	0	M1017
toy store	18.021031	-999.0	0	0.0	0	M1017
toy store	8.004458	-999.0	1	1.0	0	M1017
toy store	15.025859	-999.0	2	2.0	0	M1017
toy store	35.436027	-999.0	3	3.0	0	M1017
clothing store	6.483371	8.0	0	0.0	0	M1017
clothing store	35.618037	3.0	1	1.0	0	M1017
clothing store	35.469029	8.0	2	2.0	0	M1017
clothing store	42.952457	3.0	3	3.0	0	M1017
cafe	13.236757	-999.0	0	1.0	0	M1017
cafe	17.375994	-999.0	1	3.0	0	M1017
gym	32.809876	3.0	0	1.0	0	M1017
gym	15.942912	3.0	1	3.0	0	M1017
pet store	16.986972	-999.0	0	1.0	0	M1017
pet store	16.725956	-999.0	1	2.0	0	M1017
jewelry store	13.247758	8.0	0	2.0	0	M1017
craft shop	16.702955	-999.0	0	3.0	0	M1017

Total rows: 48

Unique stores: 17

Conditions: [8. 3. -999.]

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2.2 Spatial Trajectory Visualization

Rationale: The virtual environment positions are recorded as continuous `player_position` events (y, z coordinates in the game world). Visualizing the navigation path for each trial provides a sanity check that:

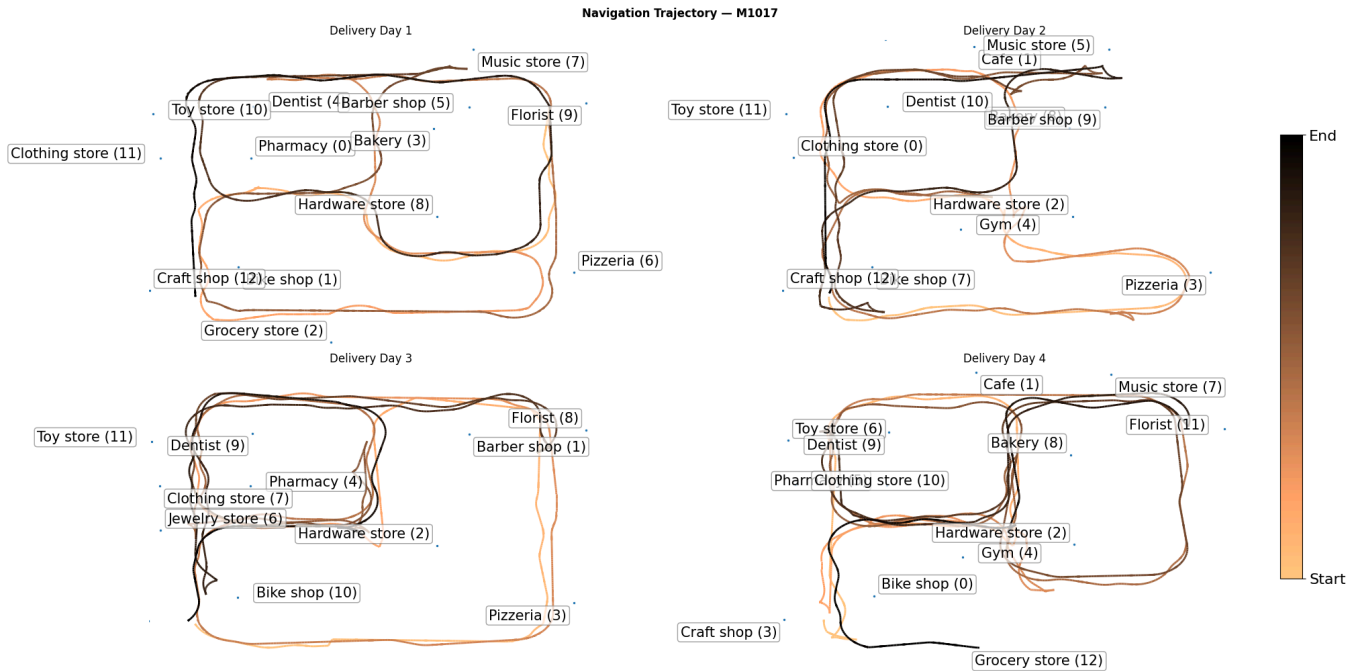
- The patient is actively navigating (not sitting still)
- Store locations are correctly labeled
- The trajectory has a plausible structure (e.g., all stores are visited in the correct order)

Implementation details:

- `extract_player_positions()` parses the comma-separated position string into numeric (y, z) columns.
- `positions_to_segments()` converts the position array into a `LineCollection` segment array, enabling color-mapped time-ordered rendering via `matplotlib.collections.LineCollection`. The color gradient (copper_r colormap, dark → light) reflects temporal progression from trial start to end.
- `remove_stationary_segments()` filters out zero-movement frames to reduce visual clutter from pause events.
- `draw_store_markers()` overlays store locations with delivery-order annotations, using staggered offsets to avoid label overlap.

Each subplot corresponds to one delivery day (i.e., trial). An absent or stationary trajectory signals a data capture failure.

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2.3 Recall Performance

Rationale: Free recall accuracy and intrusion rates are the primary behavioral outcomes of the EFR Courier Open Loop experiment. This analysis breaks down performance by stimulation condition at both encoding and retrieval to enable a first-pass assessment of whether stimulation had any observable effect on memory within this single session.

`compute_recall_stats()` returns five objects:

- `trial_stats` — per-trial summary (list length, correct recalls, intrusions, accuracy)
- `cond_stats` — per-word table with encoding condition and whether the word was recalled
- `intrusion_df` — each intrusion event labeled with its **retrieval-time** stimulation condition (PLI and ELI are pooled, as the QC report does not distinguish intrusion type)
- `ret_cond_counts` — global count of recall responses per retrieval condition
- `trial_ret_cond_counts` — per-trial version of the above, used to determine whether a condition was active in a given trial (for the 0/0 placeholder logic)

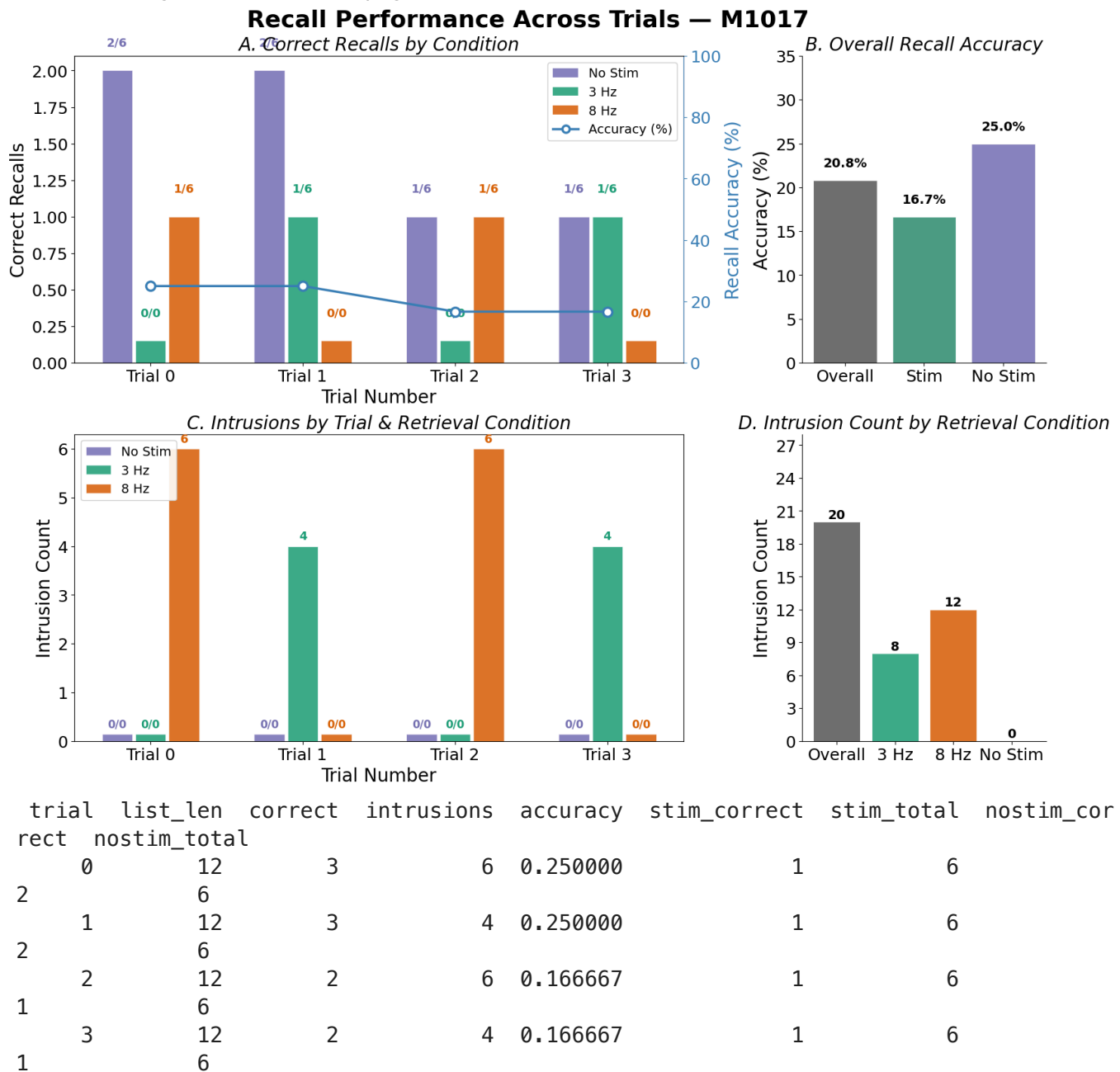
Retrieval condition assignment (`get_retrieval_condition()`): Each recall vocalization (`fr_rec_word`) is matched to the nearest `stimming` event within a ± 3000 ms window (the stimulation duration). If no stimulation event falls within that window, the recall is labeled as occurring outside stimulation (condition -999).

4-panel figure:

- **Panel A:** Correct recalls per trial, broken down by stimulation condition during encoding (3 bars/trial). The right y-axis shows overall accuracy (%), reflected by the overlaid line.
- **Panel B:** Aggregate recall accuracy across all trials, split by overall/stimulation/no-stimulation.
- **Panel C:** Intrusion count per trial, split by retrieval stimulation condition.

- **Panel D:** Total intrusion count aggregated across trials.

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Section 3 - Stimulation EEG Validation

This section validates that electrical stimulation was successfully delivered to the correct targets and that the evoked neural responses are physiologically plausible. Failed or imprecise stimulation delivery would invalidate the primary manipulation of the experiment.

3.1 Stimulated Channel Identification & Epoch Extraction

`identify_stimulated_channels()` determines which recorded bipolar channels correspond to the stimulation contacts, and partitions all channels into three groups:

Group	Definition	QC purpose
Stimulated pairs (<code>Contact_stimuli</code>)	Bipolar channels whose both leads match the <code>electrode_pos / electrode_neg</code> fields of stimulation events	Should show large stimulation artifact
Stimulated shank (<code>Channel_stimuli</code>)	Other contacts on the same electrode shank as the stimulated pair, excluding the pair itself	May show reduced artifact due to proximity; important for field-spread assessment
Non-stimulated shanks (<code>Index_other</code>)	All channels on different electrode shanks	Should show minimal artifact — serves as a reference

The matching logic uses a Counter: a channel index appears in exactly 2 of the 4 lead sets (lead1_pos, lead1_neg, lead2_pos, lead2_neg) if and only if both of its leads are part of the stimulation circuit.

`segment()` converts onset timestamps (seconds) to sample indices, constructs MNE-format events arrays, and calls `mne.Epochs` to extract -1 to +10 s windows around each stimulation event. This produces arrays of shape (n_trials, n_channels, n_samples) for each condition:

- `eeg_8HZ / eeg_3HZ` — encoding-phase stimulation epochs
- `eeg_8HZ_retrieval / eeg_3HZ_retrieval` — retrieval-phase stimulation epochs
- `eeg_no_encoding` — encoding words without stimulation (used as a reference baseline for channel QC in Section 4)

Sample rate: 1000.0 Hz
EEG data shapes:
8 Hz encoding: (12, 124, 11001)
3 Hz encoding: (12, 124, 11001)
No stimuli encoding: (29, 124, 11001)
Time axis range: -1000.0 to 10000.0 ms

3.2 Average Evoked Responses During Encoding Stimulation

Rationale: A successful stimulation should show a large artifact at the stimulated contacts, decaying artifact at neighboring shank contacts, and minimal contamination at remote channels. The spatial specificity of the artifact is an indicator that current delivery was focal and electrode impedances were appropriate.

2×3 grid figure (Figure 1): Each row corresponds to one stimulation frequency (8 Hz top, 3 Hz bottom). Each column shows the mean ± SEM voltage trace (shaded ribbon) for one channel group:

- **Non-stimulated Shanks** (blue) — reference; should be near zero with small SEM. Y-axis scaled to 1/10 of the pair amplitude
- **Stimulated Pairs** (purple) — largest artifact amplitude; y-axis scaled to peak artifact
- **Stimulated Shank** (green) — intermediate; y-axis scaled to 1/10 of the pair amplitude

The gray shaded region marks the 3-second stimulation window (0–3000 ms).

Individual trial figures (Figure 2): The 1×2 panel shows raw per-trial traces overlaid (color-mapped by trial) with the mean in black, zoomed to 0–1000 ms. Shaded grey bands mark predicted burst intervals based on the stimulation cycle structure:

- **3 Hz:** cycle = 333.3 ms, burst duration = 83.3 ms (25% duty cycle), ~9 cycles in 3 s
- **8 Hz:** cycle = 125.0 ms, burst duration = 31.3 ms (25% duty cycle), ~24 cycles in 3 s

A zoomed-in view (–20 to +50 ms) verifies that the first artifact onset begins within ± 20 ms of the stimulation marker — a key timing precision check.

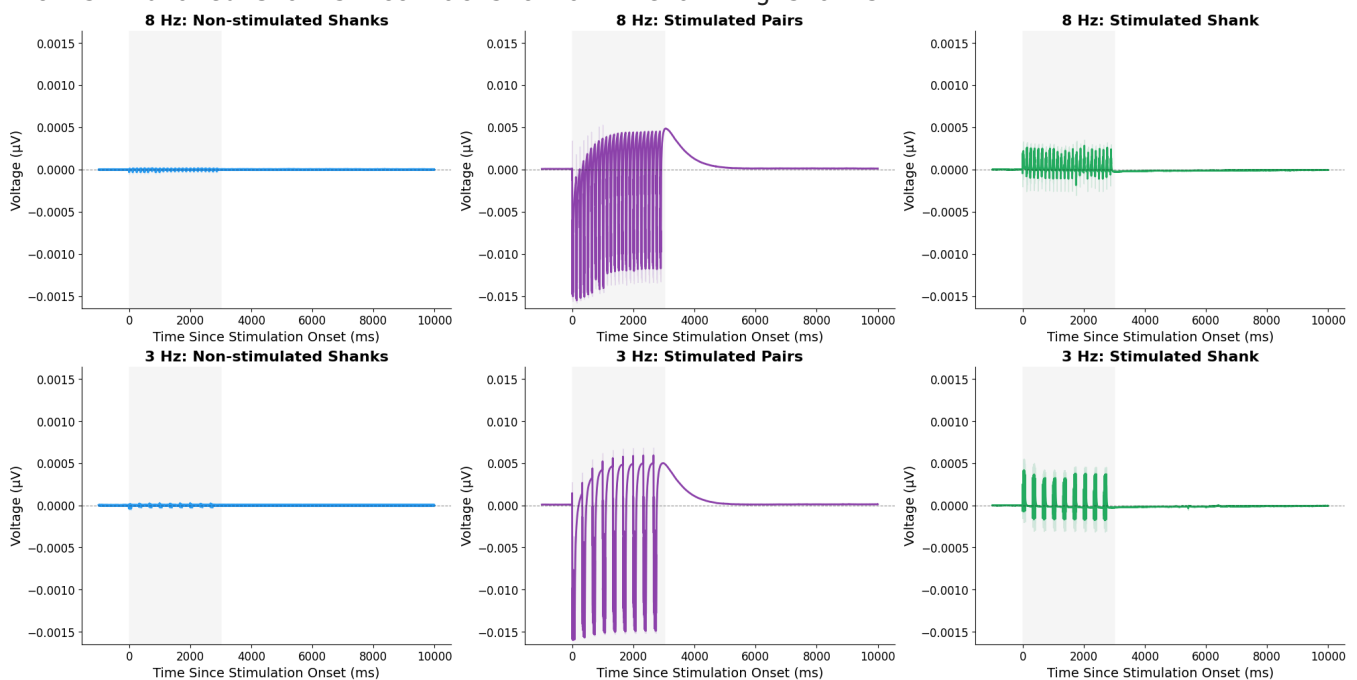
⚠ **QC flag:** If artifact onset is consistently offset by >20 ms, this indicates a synchronization error between the stimulator and the EEG recording system.

2.1 EEG Response for Stimulation Visualization during Encoding

Stimulated Pairs: the specific contacts delivering stimulation.

Stimulated Shank: all other contacts on the same electrode shank, excluding the stimulated contacts.

Non-stimulated Shanks: contacts on all remaining shanks.

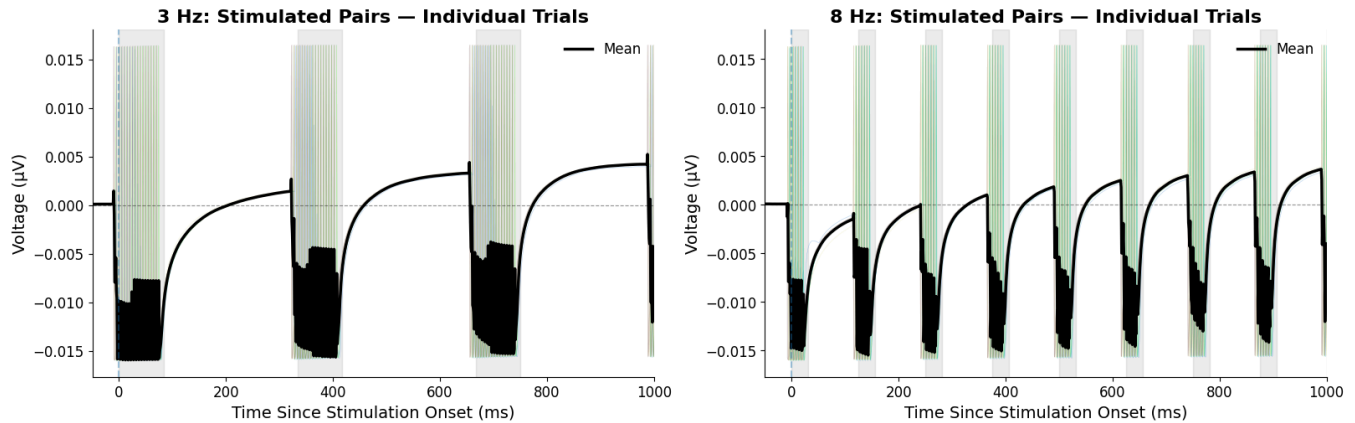


For fpulse = 3 Hz:

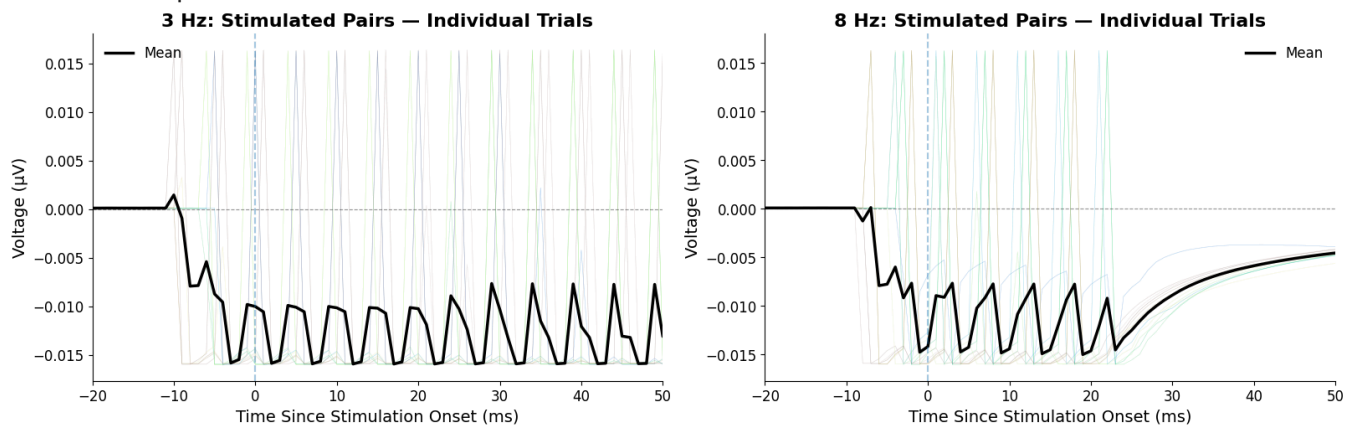
- Cycle duration: $1/3 = 333.3 \text{ ms}$
- Theta-burst duration: $0.25 \times 333.3 = 83.3 \text{ ms}$
- Total cycles in 3s: $3000 / 333.3 = 9 \text{ cycles}$

For fpulse = 8 Hz:

- Cycle duration: $1/8 = 125.0 \text{ ms}$
- Theta-burst duration: $0.25 \times 125.0 = 31.3 \text{ ms}$
- Total cycles in 3s: $3000 / 125.0 = 24 \text{ cycles}$



The first pulse onset should fall within $\pm 20 \text{ ms}$ of the stimulation onset marker.



3.3 Average Evoked Responses During Retrieval Stimulation

This cell repeats the encoding stimulation visualization (Section 3.2) for the retrieval phase, where stimulation is delivered in 3-sec on-off trains during the free-recall period. The extracted epoch window is the same as encoding (-1 to $+10 \text{ s}$), but a second shaded region at 6000 – 9000 ms marks a second stimulation burst window that should occur in the recall period.

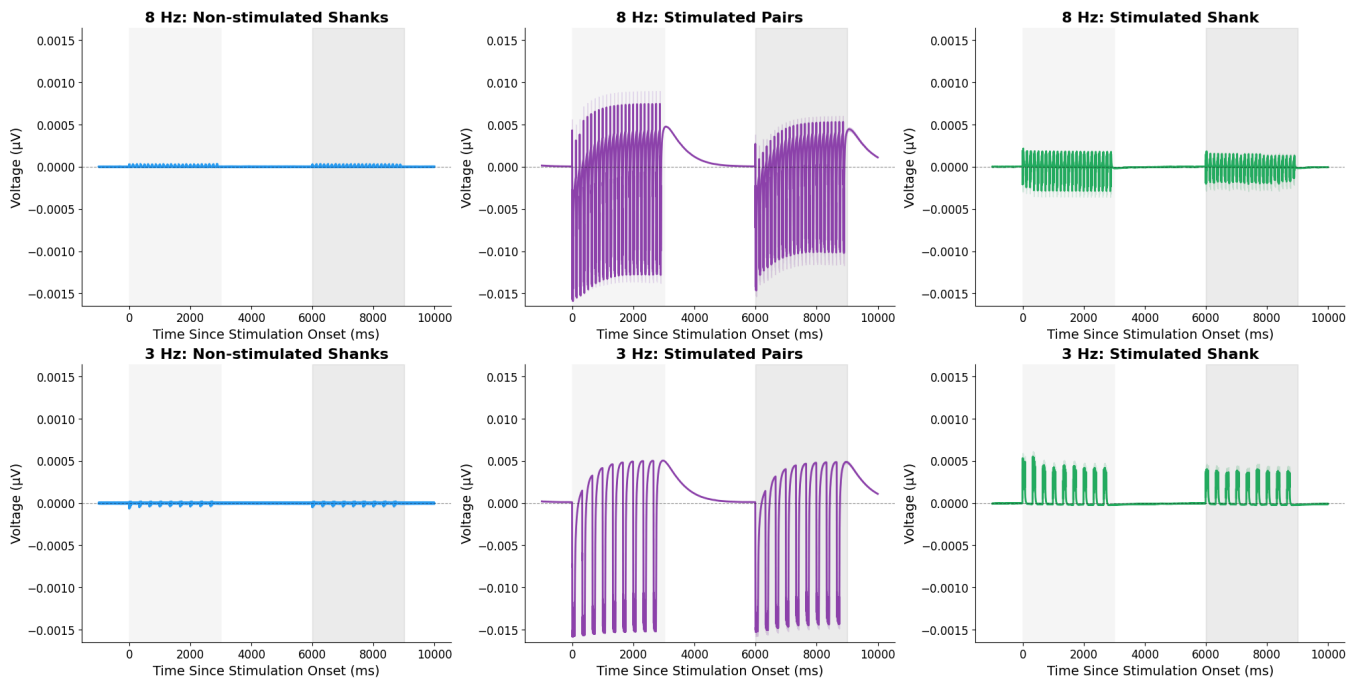
2.2 EEG Response for Stimulation Visualization during Retrival

Stimulated Pairs: the specific contacts delivering stimulation.

Stimulated Shank: all other contacts on the same electrode shank, excluding the stimulated contacts.

Non-stimulated Shanks: contacts on all remaining shanks.

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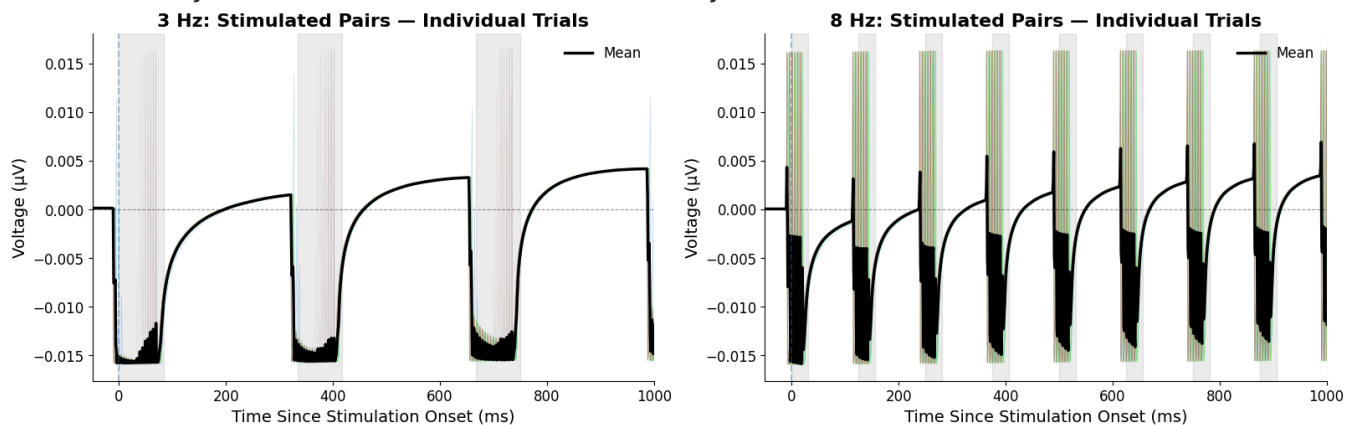


For fpulse = 3 Hz:

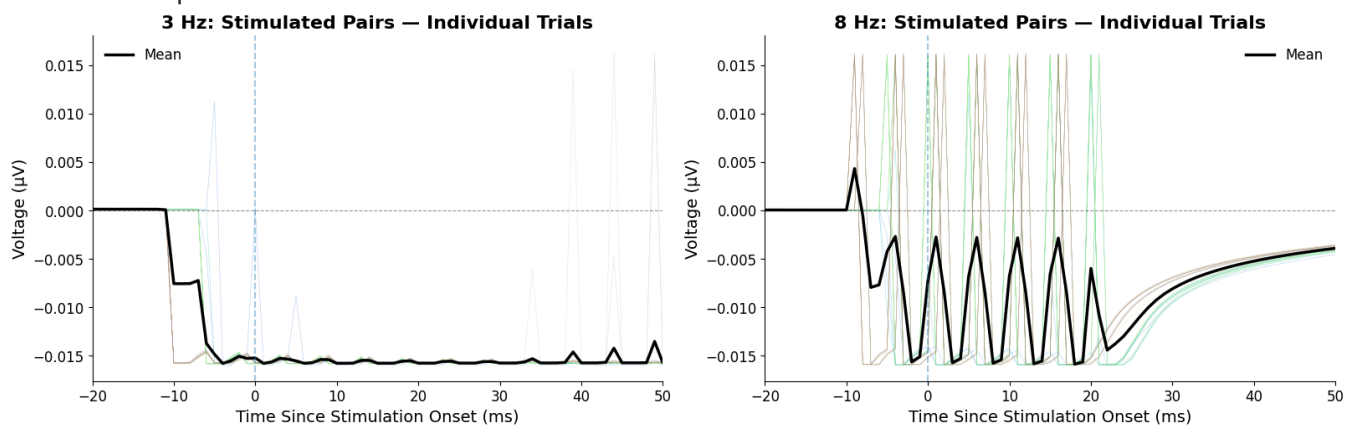
- Cycle duration: $1/3 = 333.3$ ms
- Theta-burst duration: $0.25 \times 333.3 = 83.3$ ms
- Total cycles in 3s: $3000 / 333.3 = 9$ cycles

For fpulse = 8 Hz:

- Cycle duration: $1/8 = 125.0$ ms
- Theta-burst duration: $0.25 \times 125.0 = 31.3$ ms
- Total cycles in 3s: $3000 / 125.0 = 24$ cycles



The first pulse onset should fall within ± 20 ms of the stimulation onset marker.



Section 4 — Channel Quality Control

This section implements quantitative, automated screening for bad channels using two complementary variance-based metrics applied to the no-stimulation encoding epochs (`eeg_no_encoding`). Using unstimulated epochs as input ensures that the large stimulation artifacts do not drive the outlier detection.

The channel QC module addresses the challenge of identifying recording artifacts, high-impedance contacts, and electrical interference to prevent contamination of downstream spectral or connectivity analyses.

4.1 Trial Variance Method

Principle: For each channel, compute the standard deviation of the voltage trace across trials at each timepoint, then average over timepoints. Channels with abnormally high cross-trial variability relative to the array are flagged as potentially noisy.

Procedure:

1. `trial_variability = np.std(eeg_no_encoding[:, channel, :], axis=0)` → shape `(n_channels, n_timepoints)` — for each channel and timepoint, how much does voltage vary across trials?
2. `channel_variability = np.mean(trial_variability, axis=1)` → shape `(n_channels,)` — the time-averaged cross-trial std per channel.
3. Z-score across channels; flag `|z| > 2` as outliers.

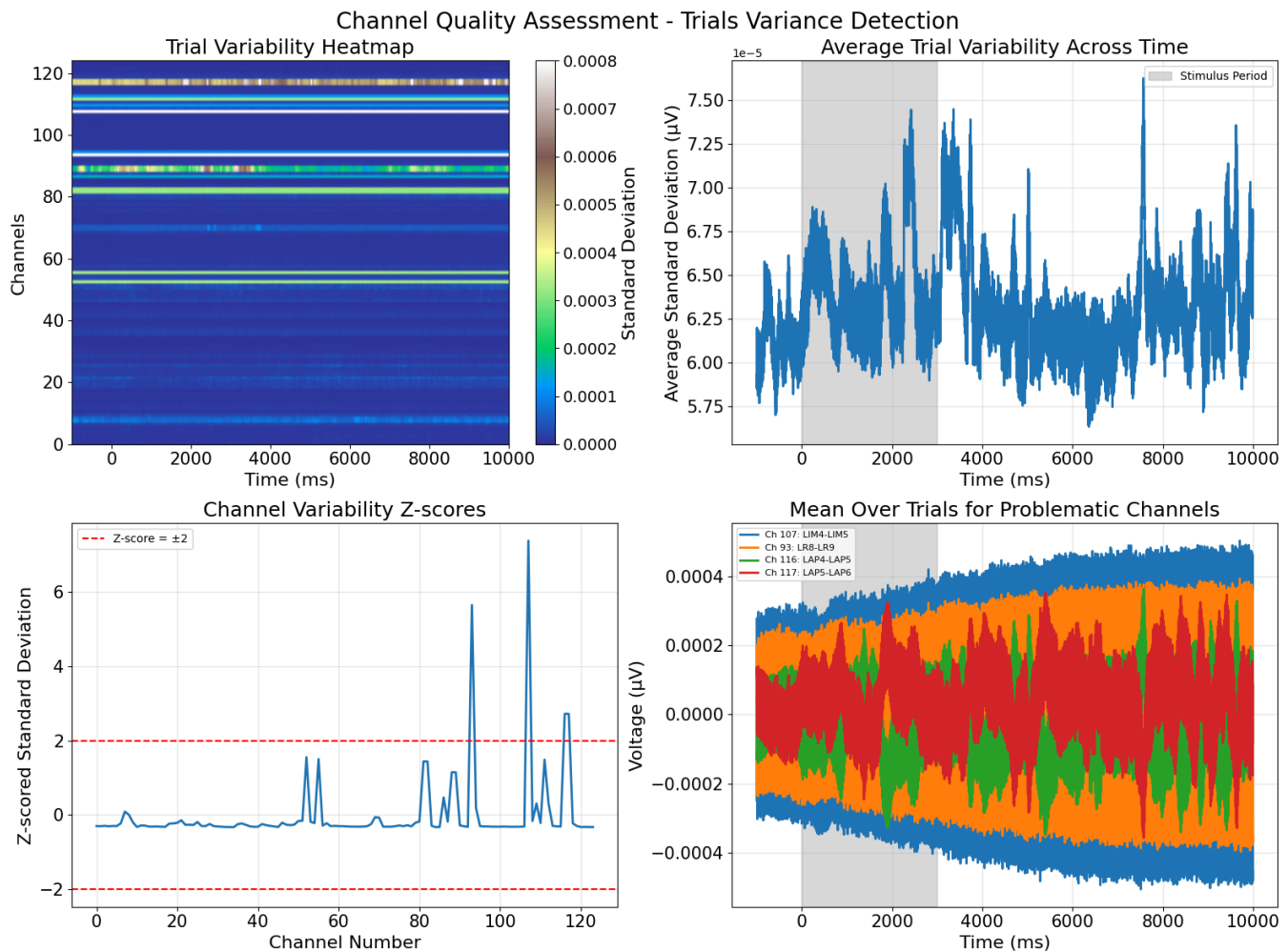
4-panel diagnostic figure:

- **Panel A (Heatmap):** Trial variability as a function of time and channel — helps visualize whether variability spikes are time-locked (suggesting an artifact aligned to task events) or channel-specific (suggesting a noisy electrode).
- **Panel B:** Mean variability across channels over time — shows whether overall noise increases during specific task phases.
- **Panel C:** Channel z-scores — the ± 2 dashed lines reflect the flagging threshold.
- **Panel D:** Trial-averaged traces for the top flagged channels — visual confirmation of whether flagged channels have drift, high-frequency noise, or burst activity.

Total channels analyzed: 124
Channels with high variability ($|z\text{-score}| > 2$): 4

Potential problematic channels:

Channel 93: LR8-LR9 (z-score: 5.65)
Channel 107: LIM4-LIM5 (z-score: 7.38)
Channel 116: LAP4-LAP5 (z-score: 2.72)
Channel 117: LAP5-LAP6 (z-score: 2.72)



4.2 Timepoint Variance Method

Principle: Rather than measuring variability *across trials* (as in 4.1), this method measures variability *across timepoints within each trial*, then averages across trials. This is sensitive to a different failure mode: channels that are internally noisy within each trial (high intra-trial temporal variance), even if they appear stable from trial to trial.

Procedure:

1. `timepoint_variance = np.var(eeg_no_encoding[:, channel, :], axis=2) → shape (n_trials, n_channels)` — within each trial, how much does voltage vary over time?

2. `trial_averaged_variance = np.mean(timepoint_variance, axis=0) → shape (n_channels,)` — average across trials.
3. Z-score and flag `|z| > 2`.

The two metrics (trial variance and timepoint variance) are **complementary**:

- High **trial variance** → inconsistent responses across trials (possibly intermittent artifact)
- High **timepoint variance** → constantly noisy within each trial (possibly broken contact, continuous artifact)

The 4-panel figure structure mirrors Section 4.1, enabling direct comparison.

Total channels analyzed: 124

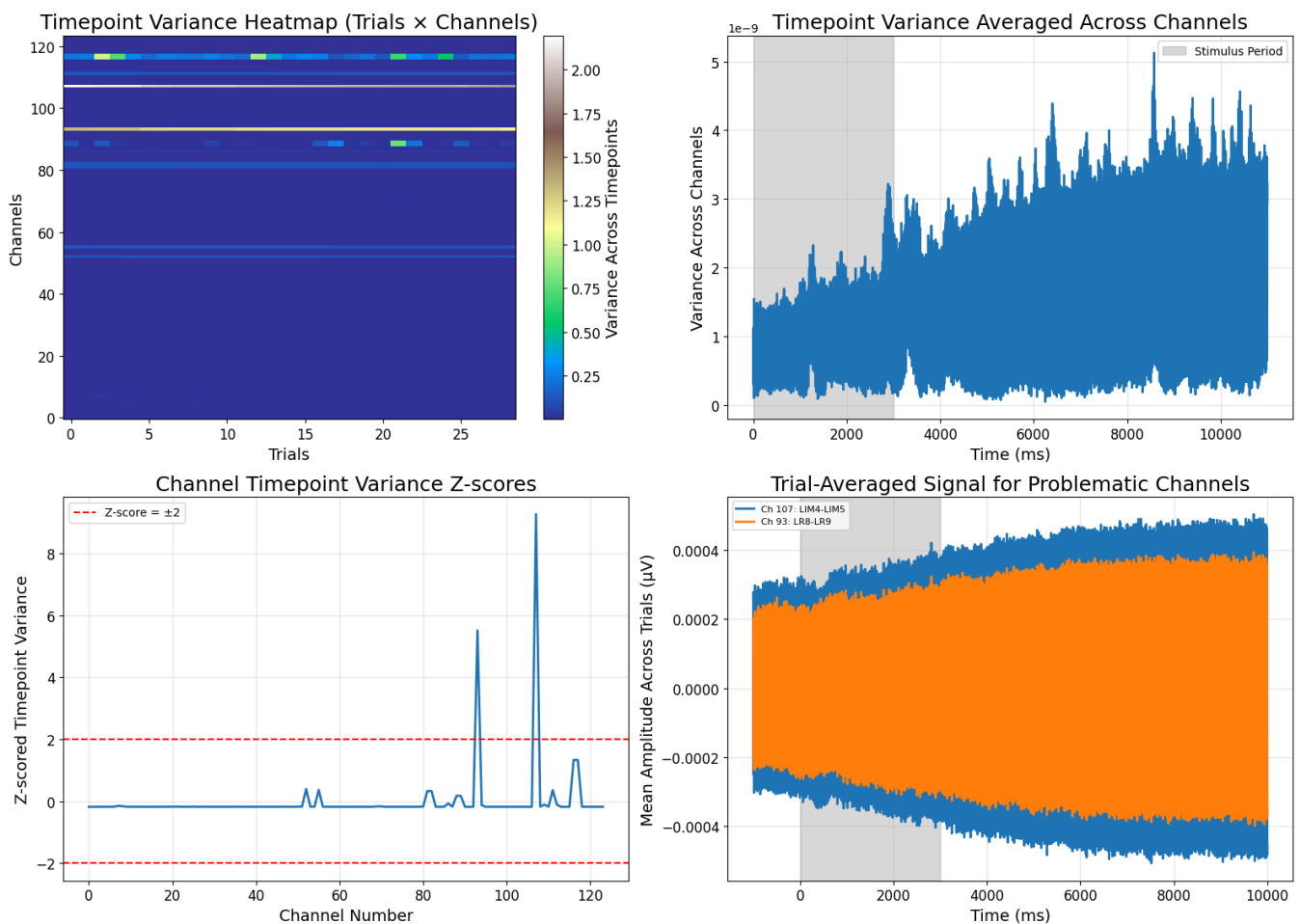
Channels with high timepoint variance ($|z\text{-score}| > 2$): 2

Potential Problematic Channels:

Channel 93: LR8-LR9 (z-score: 5.52)

Channel 107: LIM4-LIM5 (z-score: 9.27)

Channel Quality Assessment - Timepoint Variance Detection



4.3 Combined Bad-Channel Summary — All Channels Panel

This cell combines both variance metrics to produce a comprehensive per-channel summary, classifying every channel into one of four categories:

Flag	Color	Meaning
[BOTH]	Red / pink background	Flagged by both trial variance AND timepoint variance — highest confidence bad channel
[TRIAL]	Orange / orange background	Flagged only by trial variance — potentially intermittent artifact
[TIMEPOINT]	Blue / blue background	Flagged only by timepoint variance — continuously noisy but consistent across trials
(none)	Black / white background	No issues detected

The large multi-panel figure (one subplot per channel) allows rapid visual review of all channel traces simultaneously. The shared y-axis scale (`y_max = 1.2 × global_max`) enables direct amplitude comparison across channels — a channel standing out visually should match its flag status.

Usage guidance: Channels flagged `[BOTH]` are very likely to be excluded from all downstream spectral, connectivity, and decoding analyses. However, all flagged channels warrant visual review of the individual trace before exclusion. The list of excluded channels should be recorded in the session QC log.

```
Channels flagged by trial variance only (orange): [np.int64(116), np.int64(117)]
Channels flagged by timepoint variance only (blue): []
Channels flagged by BOTH (red): [np.int64(93), np.int64(107)]
```

4.4 Line Noise Screening — 55–65 Hz Band Power

Rationale: Electrical line noise at 60 Hz (US power grid) is a common artifact in hospital iEEG recordings that can contaminate spectral analyses and inflate power estimates in the gamma band. This section computes the FFT power spectrum of the trial-averaged signal for each channel and extracts mean log power in the 55–65 Hz band as a measure of line noise contamination.

Procedure:

1. Average `eeg_no_encoding` across trials: `mean_data = np.mean(eeg_no_encoding, axis=0)` → shape `(n_channels, n_samples)`.
2. For each channel, compute the one-sided FFT power spectrum and convert to dB: `10 × log10(|FFT|2 + ε)`.
3. Extract mean dB power in 55–65 Hz band.
4. Z-score across channels; channels with `z > 2` are flagged as high line noise.

Two figures:

Figure 4.4a — Band power scatter: All channels plotted as colored dots (color = electrode group) with their 55–65 Hz band power on the y-axis. Vertical red dashed lines every 32 channels aid in shank-level organization. This gives an at-a-glance view of whether line noise is uniformly distributed or concentrated in specific electrode groups.

Figure 4.4b — Per-channel power spectra: Full 0–100 Hz spectrum for every channel, with flagged channels highlighted in red and the 55–65 Hz band shaded. The Z-score is shown in each subplot title. This enables direct verification of the 60 Hz peak morphology — a line noise artifact will show a narrow peak precisely at 60 Hz, whereas broadband noise may indicate a different type of channel failure.

The z-score table (cell 13, between the two figures) provides a ranked, text-formatted report of all channels sorted by band power — convenient for copy-pasting into session notes.

4.5 Per-Trial Stimulation Artifact — Individual Trial Inspection

This section provide a trial-by-trial view of stimulation, as seen from the stimulated contact pair, zoomed to the second stimulation burst window (950–2050 ms). This is distinct from the earlier Figure 2 (Section 3.2), which showed the first burst.

Purpose: Confirm that stimulation artifacts are consistent across trials. Trial-to-trial variability in artifact amplitude or timing can indicate:

- Varying electrode impedance across the session (degrading contact)
- Hardware triggering inconsistency

Each subplot shows the contact-pair-averaged voltage for a single trial. The black dashed line at 0 ms marks stimulation onset; the red dashed line at 1000 ms marks the end of the first cycle.

Cell 15 — 3 Hz condition: `avg_eeg_stimuli_3 = mean(eeg_3HZ[:, Contact_stimuli_3, :], axis=1) → shape (n_trials, n_timepoints).`

Cell 16 — 8 Hz condition: Same procedure for the 8 Hz epochs.

⚠ **QC flag:** Look for trial that show dramatically different artifact morphology (e.g., absent artifact, reversed polarity, or saturation).

Summary Checklist

After running all modules, use this checklist to determine whether the session passes QC:

✓ Protocol Checks

- ☐ All expected event counts passed validation (n/n checks passed displayed in green)
- ☐ Stimulation was successfully delivered during all intended task phases (e.g., encoding and retrieval)
- ☐ Commanded stimulation frequencies (e.g., 3 Hz and/or 8 Hz) were correctly detected for each trial/session

✓ Behavioral Checks

- ☐ Navigation times fall within a plausible range (no extreme outliers suggesting task interruption)
- ☐ Navigation trajectories indicate active exploration of the virtual environment throughout the session
- ☐ Recall accuracy is greater than zero, indicating participant engagement with the task

✓ Stimulation verifications Checks

- ☐ Clear stimulation artifact observed at the stimulated contact pair (confirms successful stimulation delivery)
- ☐ Artifact onset occurs within ± 20 ms of the expected stimulation marker (verifies synchronization and timing accuracy)
- ☐ Artifact magnitude decays with increasing distance from the stimulated contacts (confirms spatial specificity)
- ☐ Stimulation artifact morphology is stable and consistent across trials

✓ Channel QC Checks

- ☐ List of [BOTH] -flagged channels is documented for potential exclusion from downstream analyses
- ☐ High-z noisy channels do not involve a large proportion of the array (>20% may indicate a systemic recording or referencing problem)
- ☐ Line-noise contamination is limited to a small subset of channels (widespread 60 Hz contamination may indicate grounding failure)

If any red flags are identified, notify the Herz Lab clinical/research team immediately. Early detection may allow troubleshooting, hardware adjustment, or scheduling of additional recording sessions while the patient remains hospitalized.